Review and Practice Practical

Task 1: Building the Network in Packet Tracer

1. **Launch & Setup:**
    - Open **Cisco Packet Tracer**.
    - Go to **File →\rightarrow→ Save As...** and save the project file as **HomeWiFiNetwork.pkt**.
2. **Add the Devices:**
    - **Wireless Router:** Navigate to **Network Devices →\rightarrow→ Wireless Devices**, drag the **WRT300N** (Wireless Router) onto the workspace.
    - **Three End PCs:** Navigate to **End Devices →\rightarrow→ Devices**, drag three **PC** units (PC0, PC1, PC2) onto the workspace. Rename them **PC 1**, **PC 2**, and **PC 3**.
3. **Connect the Devices:**
    - Select **Connections (Lightning Bolt icon)** →\rightarrow→ **Copper Straight-Through cable** (solid black wire).
    - Connect **PC 1** (FastEthernet0) to **Router** (Ethernet 1).
    - Connect **PC 2** (FastEthernet0) to **Router** (Ethernet 2).
    - Connect **PC 3** (FastEthernet0) to **Router** (Ethernet 3).*(Alternatively, if connecting wirelessly: Turn off the PC in the Physical tab, swap the Ethernet port for the WMP300N wireless card, turn it back on, and connect via PC Wireless)*.
4. **Configure Router LAN IP:**
    - Click on the **WRT300N Router** →\rightarrow→ **GUI** tab.
    - Under **Basic Setup →\rightarrow→ Network Setup →\rightarrow→ Router IP**:
        - **IP Address:** 192.168.10.1
        - **Subnet Mask:** 255.255.255.0
    - Scroll down and click **Save Settings**.

Task 2: Static IP Addressing Scheme (192.168.10.0/24)

Assign static IP addresses on each PC so all devices share the same subnet and point to the router's gateway:

Device Name	Interface	IPv4 Address	Subnet Mask	Default Gateway
Wireless Router (WRT300N)	LAN / Internal	192.168.10.1	255.255.255.0	*N/A (Acts as Gateway)*
PC 1	FastEthernet0	192.168.10.10	255.255.255.0	192.168.10.1
PC 2	FastEthernet0	192.168.10.20	255.255.255.0	192.168.10.1
PC 3	FastEthernet0	192.168.10.30	255.255.255.0	192.168.10.1

How to configure on each PC:

1. Click the PC →\rightarrow navigate to the **Desktop** tab.
2. Open **IP Configuration**.
3. Select **Static** (switch from DHCP).
4. Enter the corresponding **IPv4 Address**, **Subnet Mask**, and **Default Gateway** from the table above.

Task 3: Verifying Connectivity Using the ping Command

Open the **Command Prompt** tool on **PC 1** (**Desktop →\rightarrow→ Command Prompt**) and execute the following tests:

1. Ping the Router Gateway:

Plaintext

C:\> ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

2. Ping Other PCs on the Subnet:

Plaintext

C:\> ping 192.168.10.20

Reply from 192.168.10.20: bytes=32 time<1ms TTL=128

Reply from 192.168.10.20: bytes=32 time<1ms TTL=128

... Lost = 0 (0% loss)

C:\> ping 192.168.10.30

Reply from 192.168.10.30: bytes=32 time<1ms TTL=128

Reply from 192.168.10.30: bytes=32 time<1ms TTL=128

... Lost = 0 (0% loss)

- **Result:** 0% packet loss confirms that all three PCs and the wireless router communicate across Layer 2 and Layer 3.

Visual Reference Diagram

An architectural summary diagram for your project showing the topology, static IP assignments, and ping test results:

Task 4: Documentation Template (For Word / PDF Submission)

Plaintext

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                LAB REPORT: HOME WI-FI NETWORK CONFIGURATION

                    File: HomeWiFiNetwork.pkt

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1. OBJECTIVE:

To design and verify a small office/home office (SOHO) network using Cisco Packet Tracer consisting of one wireless router and three PCs on a static 192.168.10.0/24 subnet.

2. TOPOLOGY OVERVIEW:

- Central Device: Linksys WRT300N Wireless Gateway (LAN IP: 192.168.10.1/24)
- End Devices: PC 1, PC 2, PC 3 connected to Ethernet switch ports 1, 2, and 3.

3. STATIC ADDRESSING TABLE:

Device Name	IP Address	Subnet Mask	Default Gateway	Status
Router Gateway	192.168.10.1	255.255.255.0	N/A	Active
PC 1	192.168.10.10	255.255.255.0	192.168.10.1	Active
PC 2	192.168.10.20	255.255.255.0	192.168.10.1	Active
PC 3	192.168.10.30	255.255.255.0	192.168.10.1	Active

4. VERIFICATION EVIDENCE:

- Ping PC 1 -> Router (192.168.10.1): Success (0% Loss, TTL=255)

- Ping PC 1 -> PC 2 (192.168.10.20): Success (0% Loss, TTL=128)

- Ping PC 1 -> PC 3 (192.168.10.30): Success (0% Loss, TTL=128)

5. CONCLUSION:

The network topology was successfully built, statically addressed, and verified.

All devices are functioning within the same collision-free local broadcast domain.

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Save your .pkt file and submit your document alongside the project file or GitHub/Drive URL on your portal.